Experiment No. 5

Interactive Dashboard Development using Tableau / Microsoft Power BI

Aim

To design and develop an interactive business dashboard using Microsoft Power BI (the same steps apply in Tableau) for analyzing the Superstore Sales Dataset and presenting KPIs to support data-driven decision-making.

Tool and Dataset

Tool: Microsoft Power BI Desktop (Tableau Public equivalents are given in brackets). **Dataset:** Superstore Sales Dataset (Order ID, Order Date, Region, Segment, Category, Sub-Category, Customer ID, Sales, Quantity, Discount, Profit).

Note: a BI dashboard is built in the Power BI or Tableau GUI, so I cannot produce your .pbix or .twbx file. The dashboard image below is a sample layout drawn from synthetic data with the Superstore column structure, to show the target design. Build your own with the steps below and replace it with your screenshot; your numbers will differ.

Procedure

Step 1: Install the tool

Download Power BI Desktop from the Microsoft Store or the Microsoft website (Windows only). [Tableau: download Tableau Public from public.tableau.com.]

Steps 2-3: Import and verify the data

- **Home > Get Data > Text/CSV** (or Excel), select Superstore file, click **Transform Data**. [Tableau: Connect > Text file / Excel.]
- In Power Query check column names and data types: Order Date = Date, Sales/Profit/Discount = Decimal Number, Quantity = Whole Number, Region/Category/Segment = Text.
- Remove duplicates and nulls, then **Close & Apply**. For a single table there are no relationships to set; with extra tables (e.g. Returns, People) use **Model view** and join them on Order ID / Region.

Step 4: Create KPIs (DAX measures)

```
Total Sales      = SUM(Superstore[Sales])
Total Profit     = SUM(Superstore[Profit])
Profit Margin %  = DIVIDE([Total Profit], [Total Sales])
Number of Orders = DISTINCTCOUNT(Superstore[Order ID])
Avg Order Value  = DIVIDE([Total Sales], [Number of Orders])
Customer Count   = DISTINCTCOUNT(Superstore[Customer ID])
```

Place each measure in a **Card** visual (Tableau: calculated fields such as `SUM([Sales])` and `COUNTD([Order ID])` shown as text marks).

Step 5: Build the visualizations

Visual	Fields	Purpose
Clustered bar	Region (axis), Total Sales	Compare regions
Line chart	Order Date (Year/Quarter/Month), Total Sales	Sales trend, seasonality

Donut / pie	Category, Total Sales	Share of sales by category
Bar chart	Sub-Category, Total Profit	Top and loss-making products
Map (filled map)	State / Country, Total Sales	Geographic performance
Treemap	Category > Sub-Category, Total Sales	Hierarchical share of sales

Step 6: Add interactivity

- **Slicers:** add Slicer visuals for Year (Order Date), Region, Category and Segment. [Tableau: right-click field > Show Filter, then use *Apply to Worksheets > All Using This Data Source*.]
- **Drill-down:** drag Category and Sub-Category (or Year, Quarter, Month) into the Axis well, then use the drill arrows. [Tableau: create a Hierarchy.]
- **Cross-filtering:** clicking a bar or slice filters all other visuals (**Format > Edit interactions** to control it). Add **Tooltips** for extra detail.

Step 7: Design the dashboard layout

Cards (KPIs) across the top, charts in the middle, slicers along the left or top, a clear title, a consistent colour theme (View > Themes), aligned visuals, and readable fonts. The sample layout is shown below.

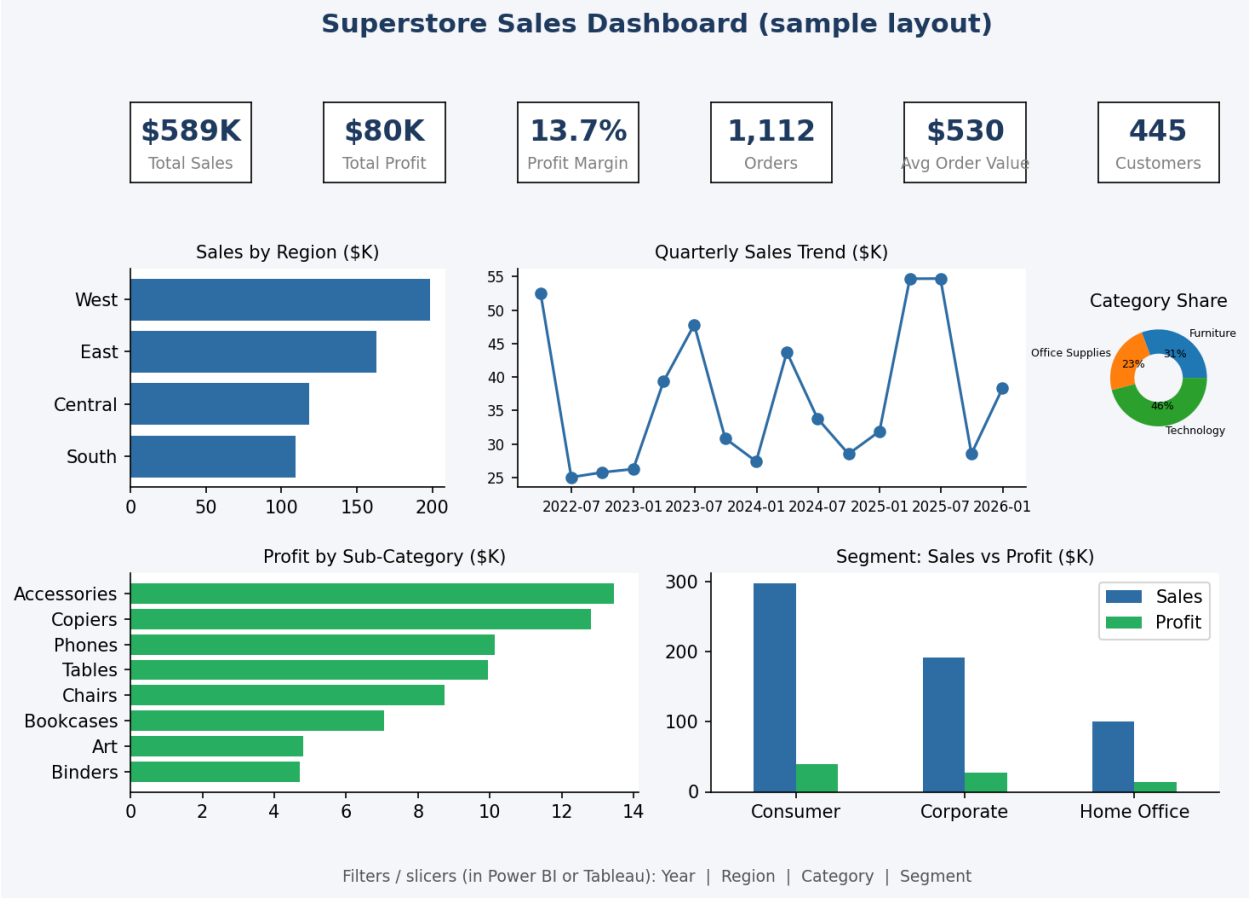


Figure: sample dashboard layout (synthetic data)

Step 8: Analyze the dashboard

Using the slicers, compare regions, categories and time periods, and note the top and bottom performers.

Step 9: Publish / export

- **Power BI:** Home > Publish to Power BI Service (needs a work/school account) or File > Export > PDF / PowerPoint.

- **Tableau Public:** File > Save to Tableau Public (publishes the dashboard as a link); or Dashboard > Export as Image / PDF.

Step 10: Report - Key Business Insights (from the sample layout)

- **KPIs:** total sales about $589K, total profit about $80K (margin about 13.7%), about 1,100 orders, average order value about $530, and about 445 customers.
- **Regions:** West has the highest sales and profit; South is lowest and needs targeted promotion.
- **Trend:** quarterly sales rise over time with seasonal peaks, so stock and staffing should be planned for them.
- **Products:** Technology items (Accessories, Copiers, Phones) are the largest profit contributors; Office Supplies items give the least profit.
- **Segments:** Consumer is the largest segment, followed by Corporate and Home Office; Corporate and Home Office can be grown with bulk-order offers.
- **Areas for improvement:** improve the South region, control discounts that reduce margin, and promote high-margin categories. *Update these with the findings from your own dashboard.*

Conclusion

An interactive dashboard with KPIs, multiple charts, slicers and drill-down was designed for Superstore data. It allows managers to explore performance by region, category, segment and time, identify strengths and weak areas, and take data-driven decisions.

Answers to Questions

Q1. What is Business Intelligence (BI)? How does it support organizational decision-making?

Business Intelligence is the set of technologies, tools and processes used to collect, integrate, analyze and present business data so that it becomes useful information. A typical BI process is: data sources (databases, ERP, CRM, files) → ETL / data warehouse → analysis → reports and dashboards.

- Gives a single, reliable view of business performance (one version of the truth).
- Identifies trends, problems and opportunities early.
- Supports forecasting, budgeting and strategy using facts instead of guesses.
- Saves time through automated reporting and self-service analytics.

Q2. What is an interactive dashboard? How is it different from a static report?

An interactive dashboard is a single-screen visual display of KPIs and charts that users can filter, click, drill into and refresh to explore the data themselves.

Aspect	Interactive dashboard	Static report
Interaction	Filters, slicers, drill-down, tooltips	None; fixed view (PDF, printout)
Data	Live or regularly refreshed	Snapshot at the time of creation
Questions answered	Many, explored by the user	Only the ones the author planned
Audience	Managers exploring different angles	Formal records, compliance, archives
Update effort	Automatic refresh	Must be rebuilt/re-issued

Q3. Compare Tableau and Microsoft Power BI based on features, usability, and business applications.

Aspect	Tableau	Power BI
Features	Very strong, flexible, highly customized visualizations; powerful for complex exploration	Wide range of visuals, DAX for calculations, AI visuals, Power Query for ETL
Usability	Drag-and-drop, but steeper learning curve for advanced work	Familiar Excel-like interface; easier for beginners
Cost	Higher licence cost (free Tableau Public with public data)	Lower cost; Power BI Desktop is free
Integration	Connects to many sources; owned by Salesforce	Deep integration with Excel, Azure, Teams and the Microsoft ecosystem
Data handling	Good for large data and visual analytics	Good for medium to large data via modelling and DirectQuery
Best for	Data analysts, visual storytelling, research and consulting	Enterprises using Microsoft tools, self-service BI, cost-sensitive teams

Q4. What are Key Performance Indicators (KPIs)? Give examples of KPIs used in sales analytics.

KPIs are measurable values that show how effectively an organization is achieving its key business objectives. A good KPI is specific, measurable, relevant and time-bound. Examples in sales analytics:

- Total Sales (revenue) and Total Profit
- Profit Margin %

- Number of Orders and Average Order Value
- Customer Count, Customer Acquisition Cost and Customer Retention Rate
- Sales Growth % (month on month / year on year)
- Conversion Rate, Sales Target Achievement, Return Rate

Q5. Explain the purpose of filters, slicers, and drill-down functionality in dashboards.

- **Filters:** restrict the data shown by a condition (e.g. only 2024, only West region) at the visual, page or report level.
- **Slicers:** visible on-screen filter controls (buttons, dropdowns, date sliders) that let any user change what all connected visuals show; they make filtering easy and obvious.
- **Drill-down:** moving from summary to detail along a hierarchy (Year → Quarter → Month, or Category → Sub-Category → Product) to find the cause of a result.

Together they turn a single dashboard into many views and let users answer their own follow-up questions.

Q6. What factors should be considered while designing an effective business dashboard?

- **Audience and objective:** know who will use it and which decisions it supports.
- **Right KPIs:** only the metrics that matter, with targets or comparisons.
- **Appropriate chart types:** bar for comparison, line for trend, map for geography, etc.
- **Layout:** most important KPIs at the top-left, logical grouping, adequate white space, no clutter (about 5-7 visuals).
- **Consistent colours and fonts**, with colour used meaningfully (red/green for bad/good) and accessible for colour-blind users.
- **Interactivity:** slicers, drill-down and tooltips.
- **Performance and data quality:** fast loading, accurate and refreshed data.
- **Clear titles and labels** with units, and mobile/screen-size suitability.

Q7. How can dashboards help managers monitor organizational performance in real time?

Dashboards connected to live or frequently refreshed data (DirectQuery, streaming datasets, scheduled refresh) display current KPIs such as daily sales, stock levels or website orders. Managers can compare actual figures with targets, see colour-coded alerts, and set data alerts that notify them when a metric crosses a threshold. This lets them detect problems (a sales drop in a region, stock-outs) immediately, act quickly, and avoid waiting for monthly reports.

Q8. Why is data visualization important in Business Intelligence tools?

- Visuals are understood faster than tables of numbers.
- Patterns, trends, outliers and comparisons become obvious.
- Non-technical decision-makers can use insights without analytical skills.
- They support storytelling and persuasion in meetings.
- They make errors and anomalies easy to spot, and provide a base for interactive exploration.

Q9. What insights can be derived from a sales dashboard for an e-commerce company?

- Total revenue, profit, orders and average order value, and their growth over time.
- Best-selling and most profitable products or categories, and slow-moving stock.
- Best and worst regions or cities, and delivery performance by area.
- Customer segments, new versus repeat customers, and customer lifetime value.
- Seasonality and campaign effects (festival sales, discount periods).
- Impact of discounts on sales and profit; return and cancellation rates.
- Traffic sources and conversion rate, showing which marketing channels work.

Q10. Explain how interactive dashboards improve communication between data analysts and business stakeholders.

Analysts speak in data and models; stakeholders think in business questions. An interactive dashboard acts as a common visual language between them. Stakeholders can explore the data themselves, so fewer ad-hoc report requests are needed. Insights are presented in simple visuals instead of technical tables, and everyone looks at the same numbers (one source of truth). Feedback is faster, as stakeholders can apply filters and ask better questions during a meeting, and analysts can refine the dashboard. This increases trust in the data and speeds up decisions.

Result: An interactive business dashboard with KPIs, visualizations, slicers and drill-down was designed, and key business insights were summarized.